

Polynomial Long Division

Dividing polynomials by long division

Directions

- Set out every division in the long-division frame. Divide the leading terms, multiply back, subtract, bring down — and subtract the whole line, not just its first term.
- Before you start, write the dividend with *every* descending power present. A missing power gets a 0 placeholder.
- Report the answer as $\text{quotient} + \frac{\text{remainder}}{\text{divisor}}$, or as the quotient alone when the remainder is 0.

1. Divide: $\frac{x^2 + 7x + 12}{x + 3}$

Answer: _____

2. Divide: $\frac{x^2 + 5x + 9}{x + 2}$

Answer: _____

3. Divide: $\frac{2x^3 - 3x^2 - 11x + 6}{x - 3}$

Answer: _____

4. Divide: $\frac{x^3 - 8}{x - 2}$

Write the dividend as $x^3 + 0x^2 + 0x - 8$ before you begin.

Answer: _____

5. Divide: $\frac{3x^2 - 5x + 4}{x - 1}$

Answer: _____

6. Divide: $\frac{x^4 - 5x^2 + 4}{x^2 - 1}$

Answer: _____

7. Consider $\frac{x^3 + 2x^2 - 5x + 1}{x - 2}$.

(a) Divide, and give the quotient with its remainder.

Answer: _____

(b) Now substitute $x = 2$ into the dividend $x^3 + 2x^2 - 5x + 1$. What value do you get?

Answer: _____

8. Divide: $\frac{2x^3 + x^2 - 8x + 5}{x + 3}$

Answer: _____

9. Divide: $\frac{x^3 + 4x^2 - 3x + 2}{x^2 + 1}$

The divisor is a quadratic, so stop when what is left has degree below 2.

Answer: _____

10. Divide: $\frac{2x^4 - 3x^3 + x - 6}{x - 2}$

Answer: _____

11. Consider $\frac{x^3 - 7x + 6}{x - 1}$.

(a) Divide. The remainder is 0, so give the quotient.

Answer: _____

(b) Use part (a) to factor $x^3 - 7x + 6$ completely.

Answer: _____

12. The division $\frac{x^3 + kx^2 - 4x + 12}{x - 2}$ leaves no remainder.

(a) Find the value of k .

Answer: _____

(b) Explain how the remainder theorem lets you find k without carrying out the long division.

Answer: _____